

— what was watched, skipped or rewatched. This behavioral dataset is a data product. In open source ecosystems, platforms like MovieLens provide public datasets for similar purposes. These datasets power models and experiments in academic and community projects. Colleges use placement dashboards to track student information, which includes GPA scores, internship experience and test results. The cleaned and visualised data becomes an interactive product through open source tools such as Superset or Metabase, which benefits both recruiters and students. The coding platforms HackerRank and LeetCode track student performance by collecting problem-solving data. The open source edtech platform Moodle generates learning analytics data products, which deliver personalised feedback to students. Student projects that use Arduino or Raspberry Pi to develop an IoT based health tracker produce valuable time series data. The combination of structured data with visualisation tools like Grafana or Plotly transforms information into a usable product for patients and doctors to monitor their health progression.

According to Valtech, a global digital consultancy headquartered in Paris, applying DaaP principles led to a 12% improvement in forecasting and an error reduction worth \$2.7 billion. This is because reliable, discoverable data allows teams to move faster.

Data monetisation: Direct and indirect strategies

Modern companies elicit value from data using two primary methods — direct and indirect monetisation. Direct monetisation involves selling or granting licences to use data with other organisations. Businesses can provide data in its original state or can offer processed data, which combines multiple datasets after cleaning and enrichment. The data becomes more valuable to potential buyers when it is processed.

Aspect	Traditional data	Data as a Product
Accessibility	Limited to IT	Open to analysts, engineers, teams
Maintenance	Ad hoc updates	Continuously maintained
Documentation	Often missing	Standardised and user focused
User experience	Backend focused	Designed for discovery
Purpose	Storage and reporting	Decision making, automation, ML training

DaaP versus traditional data

Data products are available for purchase through various pricing models.

The subscription model requires customers to pay a continuous fee to access the data. The data licensing model provides permanent or time-limited access to data at a fixed price. The customer must pay for data usage under the usage or consumption-based pricing model.

Open source platforms simplify the process of data preparation and distribution for direct monetisation. For example, CKAN (Comprehensive Knowledge Archive Network) operates as an open source data management system that helps governments and companies share their datasets. Platforms like data.world and Open Data Hub offer structured, searchable datasets, which become monetisable through API integration and visualisation tools. The open datasets found on GitHub and Kaggle often begin as public resources before companies refine them for commercial sale through open licences or business terms.

Multiple industries participate in direct data sales. Credit bureaus provide banks with credit scores together with repayment history information. Logistics companies market delivery and traffic information to urban planning agencies. The financial services leader Bloomberg generates billions of dollars annually through the sale of real-time data feeds to worldwide traders and analysts.

Indirect monetisation involves using business data to enhance products and services, and make better internal decisions. A common

example is personalisation. Spotify uses user behavioural data to generate recommendations about audio and video content. The system enhances user interaction while maintaining their subscription status. Sensor data collected from machines in factories enables predictive maintenance, which decreases operational expenses. The user data collected by tech giants such as Google enables them to show highly targeted advertisements. This generates huge revenue. According to Statista, Google's ad segment produced more than \$260 billion in revenue during 2024.

Open source tools play an important role in this strategy. Take Prometheus, for example. It collects time series metrics in real time. This helps engineering teams monitor their infrastructure and set up automated alerts. Apache Flink processes large-scale event streams, making it perfect for real-time recommendations or detecting anomalies. These tools are key to turning raw data into valuable insights that can improve internal operations or enhance customer experience. Amazon uses purchase history data to generate suggestions that really boost cross-selling. A smart home system that automates lighting based on usage data is another good example to monetise data by increasing the product's value. Students working on such systems with open source tools like Home Assistant or Node RED are already putting DaaP principles into action.

Here's a quick comparison of both strategies: